Lecture Exam Statistics for AI SS2021 Group 2

Question 1

1) A decorator has a stock of 50 cans of paint. 10 cans contain white paint, 5 cans blue paint, 8 cans green paint, 3 cans yellow paint, 6 cans red paint, 10 cans orange paint, and 8 cans grey paint. If he chooses a can at random, what is the probability that this can contains blue paint or green paint or orange paint? [10 points]

5 + 8 + 10 = 23

23/ 50 = 0.46 = 46 %

Question 2

- 2a) Why plays the "normal distribution" such an important role in inductive statistics? [4 points]
- 2b) Explain with a few words, what is meant by a "type I error" and a "type II error" in inductive statistics. [4 points]
- 2c) Explain with a few words, what is understood under the term "p-value" in inductive statistics. [4 points]
- 2d) Explain with a few words, the difference between a metric-continuous and a metric-discrete attribute in descriptive statistics. State one example for each kind of these attributes. [4 points]
- 2e) Describe with a few words the advantage and disadvantage of the "arithmetic mean" and "mode" (= modal value) as measures of location. [4 points]

2a) the central limit theorem tells us that a population with high smaple size is normal distributed, many things in "Nature" are normaly distributed (IQ, height of humans) and other distributions can be transformed into a normal distribution

2b) a type 1 error is the rejection of a true null hypothesis ("false positive" finding) ; example : an innocent person is convicted

a type 2 error is the non-rejection of a false null hypothesis ("false negative" finding) ; example : a guilty person is not convicted

2c) The p-value is the probability - under the null hypothesis - of obtaining a "result" (The p-value is the concrete type 1 error of a data situation.)

If the p-value is high, there is a high likelihood of obtaining said result under the null hypothesis therefore the null hypothesis is not rejected. If the p-value is small --> the null hypothesis is rejected

2d) metric discrete data = values from a fixed list of numbers (=counting apples on a tree)

metric continous data = values can be determined with infinite precision (= measuring the weigth of an apple in gram -> theoretically the weight can be measured with an infinite precision)

2e) "arithmetic mean"= disadvantages = can only be used for metric data, not stable against great outliers in the data advantages = if the data has no great outlieres, it is a great measure of location (we get the middle of the data)

"mode" = disadvantage = must not be unique advantage = we get the attribute with the highest frequency

Question 3

3) The police plans to enforce speed limits by using radar traps at three different locations within the city limits. The radar traps at each of the locations L1, L2 ,and L3 will be operated 40%, 30% ,and 50% of the time. If a person who is speeding on her way to work has probabilities of 20%, 10%, 70%, respectively, of passing through these locations, what is the probability that she will receive a speeding ticket? [10 points]

40 * 0.2 = 8

30 * 0.1 = 3

50 * 0.7 = 35

= in total = 46 --> 46 %

The probability that she will receive a speeding ticket is 46 %

Question 4

4) Someone has a population of 100 guinea pigs and has calculated the following information about the age in months X of these guinea pigs.

$$\sum_{i=1}^{100} x_i = 12776$$

$$\sum_{i=1}^{100} (x_i - \bar{x})^2 = 54534$$

Furthermore, it's known that these guinea pigs weigh on average 1106 grams with a standard deviation of 43 grams and that the covariance Cov(x,y) between age and weight is 753.119.

- a) Calculate the arithmetic mean of this population of 100 guinea pigs. [5 points]
- b) Calculate the variance and standard deviation of this population of 100 guinea pigs. [8 points]
- c) Estimate the weight of a guinea pig which is 37 months old based on a linear regression model using the provided information. [15 points]

a) (1/100) * 12776 = 127.76

b) Var = (1/100) * 54534 = 545.34 standard deviation = 23.3525

c) CoVar = 753.119

Var = 1849

b = CocVar/ Var(x) = 753.119/ 545.34 = 1.381

a = y - (b * x) = 1106 - (1.381 * 127.76) = 929.56344

yi = a + b * xi = 929.56344 + 1.381 * 37 = 980.6644 grams

Question 5

5) On behalf of the owner of a winery, the normally distributed average bottled quantity of wine, which is bottled in 750 ml wine bottles, should be investigated. 10 bottles are randomly selected, and the filling quantities of these bottles are checked and the following results were calculated based on this sample.

$$\bar{x} = 749.8ml \qquad s = 4.97ml$$

- a) Use an appropriate statistical test, to verify if there is a statistically significant difference from the reference value 750 ml which the filled wine bottles should have. State appropriate statistical hypotheses, the test statistic, the critical value and the test decision (type I error α = 5%, data is normally distributed). [10 points]
- b) The owner of this winery needs cork plugs to close the wine bottles. Based on a sample of 1000 cork plugs, he found out that 10% of these cork plugs were defective and cannot be used to close the wine bottles. Calculate based on this sample a 95% confidence interval for the true unknown percentage of defective cork plugs. [8 points]
- c) Someone else has performed a "two sample t-test" (type I error α = 5%) on the research question "Is there a difference between group X and group Y regarding mean cholesterol levels?" and got a p-value of 74.1% What is the test decision? [4 points]

a)

$H_{0:}$ u= 750 ml

$H_{1:}$ u $_{\text{not-equal}}$ = 750 ml (wirte "not-equal" because don't know how to make it here otherwise)

$t_{9,0,975}$= critical value = 2.26

test = ((749.8-750) / 4.97) * sqrt(10) = - 0.127255

|-0.127255| < 2,26 --> H_0 cannot be rejected

b) 8.14% - 11.86%

c) The null hypothesis is not rejected

Question 6

6) A small bicycle rental shop has only 10 bicycles available for rent. According to previous experience with a probability of 80% one bicycle is rented. Furthermore, it can be assumed that the individual bicycles are rented independently of each other. Calculate the probability that at most 9 of the bicycles are rented. [10 points]

k = 9

n = 10

p = 80 % for one cycle

result = 0.892626 = 89.263 %